Newton's first law and inertia

LEARNING TARGET · R6 I can explain Newton's 1st Law in my own words — and why 2026-XJ keeps moving.

A · KEY VOCABULARY

Force	A push or pull from an interaction with another object.
Net force	The combined effect of all forces on one object.
Newton's 1st law	*1st law:* at rest stays at rest; moving keeps its velocity — unless $\Sigma F \neq 0$.
Inertia	An object keeps its velocity unless a net force changes it. More mass, more inertia.

B · KEY IDEA + UNITS

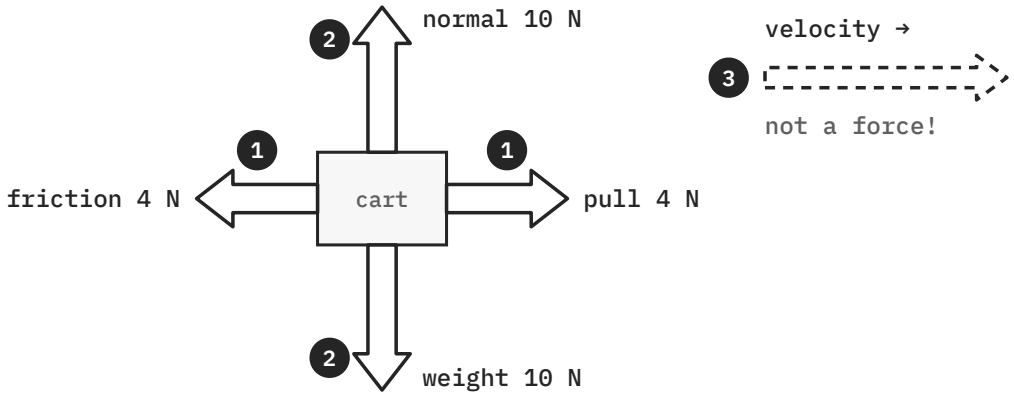
$\Sigma F = 0$
→ $a = 0$
→ velocity stays the same

in an inertial frame (not speeding up or turning)

`force` newtons (N)
`mass` kilograms (kg)

C · COLOR IT · BALANCED, STILL MOVING

A cart rolls right; forces balance.



- ☐ **1 · ORANGE** Horizontal forces (balanced)
- ☐ **2 · ORANGE** Vertical forces (balanced)
- ☐ **3 · PURPLE** Velocity arrow (not a force)

D · WORKED EXAMPLE

A cart rolls right, then slows down. What does that tell you about its net horizontal force?

GIVEN	EQUATION	WORK	ANSWER
moving right speed decreasing	$a \neq 0$ → net force $\neq 0$	velocity changes → there is a net force; it opposes the motion	**Net force left** opposite the motion

E · THE 2026-XJ THREAD

DAY 9 OF 22 · NEW TOOL: INERTIA

Out in space, nothing pushes XJ along. It keeps moving because of inertia, so no force is needed to explain its motion — only to explain a change in it.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else